

Cerinte Laboratorul 11

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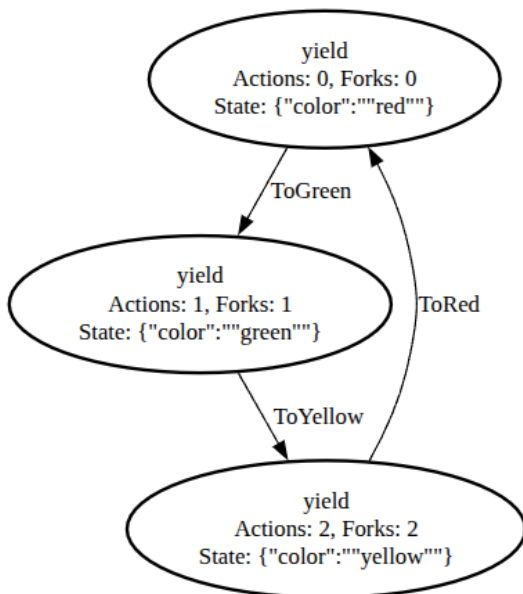
Grupa 263

Ex 1

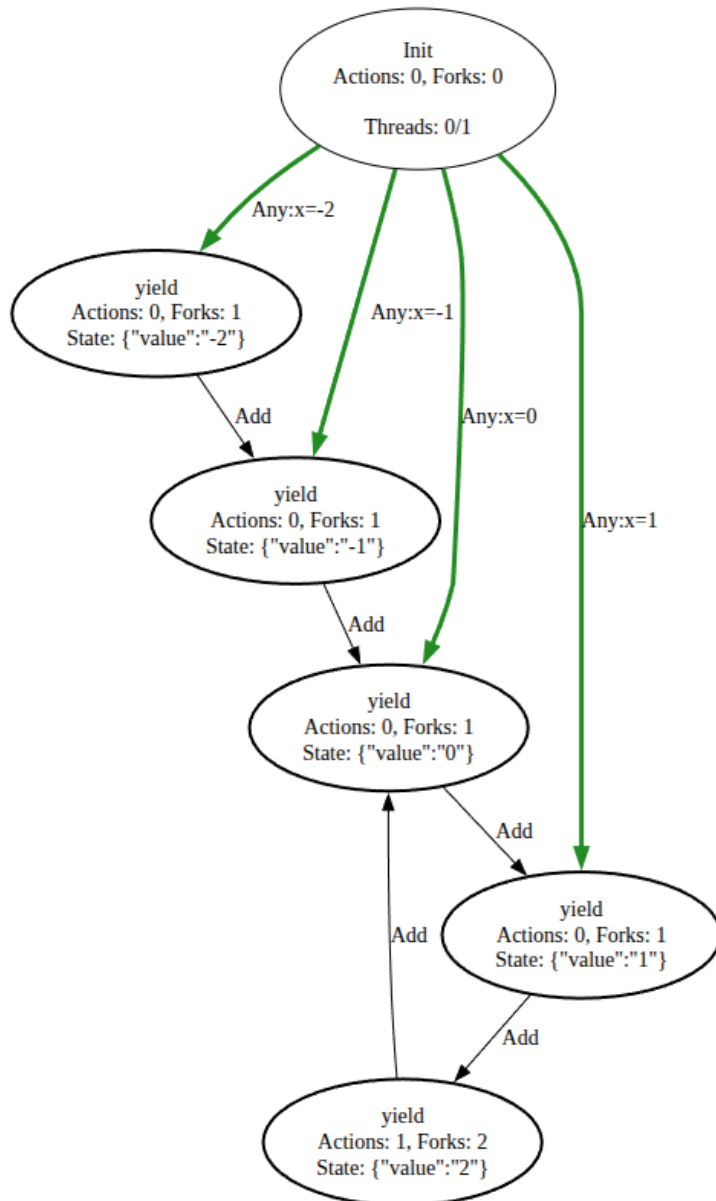
```
1 action Init:
2   a = 0
3   b = 0
4
5 action Add:
6   oneof:
7     a = (a + 1) % 3
8     b = (b + 1) % 3
9
```

Output [States Graph](#)

Model checking BuzzyMcBee.json
configFileName: fizz.yaml
StateSpaceOptions: options:{max_actions:100 max_concurrent_actions:2 crash_on_y
Nodes: 18, queued: 0, elapsed: 32.412259ms
Time taken for model checking: 32.454771ms
Valid Nodes: 18 Unique states: 9
IsLive: true
Time taken to check liveness: 137.163989ms
PASSED: Model checker completed successfully



Ex 2



```

always assertion AlwaysLessThan3:
    return value <= 2

action Init:
    value = 0
    any x in range(-2, 2):
        value = x

atomic action Add:
    # Aici am schimbat din < 3 in < 2 pentru a preveni
    # ca valoarea sa ajunga vreodata la 3.
    if value < 2:
        value += 1
    else:
        value = 0

```

Output [States Graph](#)

```

Model checking BuzzyMcBee.json
configFileName: fizz.yaml
StateSpaceOptions: options:{max_actions:100 max_concurrent_actions:2 crash_on_y
Nodes: 6, queued: 0, elapsed: 88.8977ms
Time taken for model checking: 88.939069ms
Valid Nodes: 6 Unique states: 5
IsLive: true
Time taken to check liveness: 121.911954ms
PASSED: Model checker completed successfully

```

Ex 3

```

MAX_HOUR=12
action Init:
    hour = MAX_HOUR
    meridiem = "am"

atomic action HourChange:
    # Mutam logica aici: cand trece de la 11 la 12, se schimba meridiem-ul
    if hour == MAX_HOUR - 1:
        meridiem = "pm" if meridiem == "am" else "am"

    hour = (hour % MAX_HOUR) + 1

transition assertion ClockBehavior(before, after):
    if before.meridiem != after.meridiem:
        return before.hour == MAX_HOUR - 1 and after.hour == MAX_HOUR
    else:
        return (before.hour == 12 and after.hour == 1) or (after.hour == be

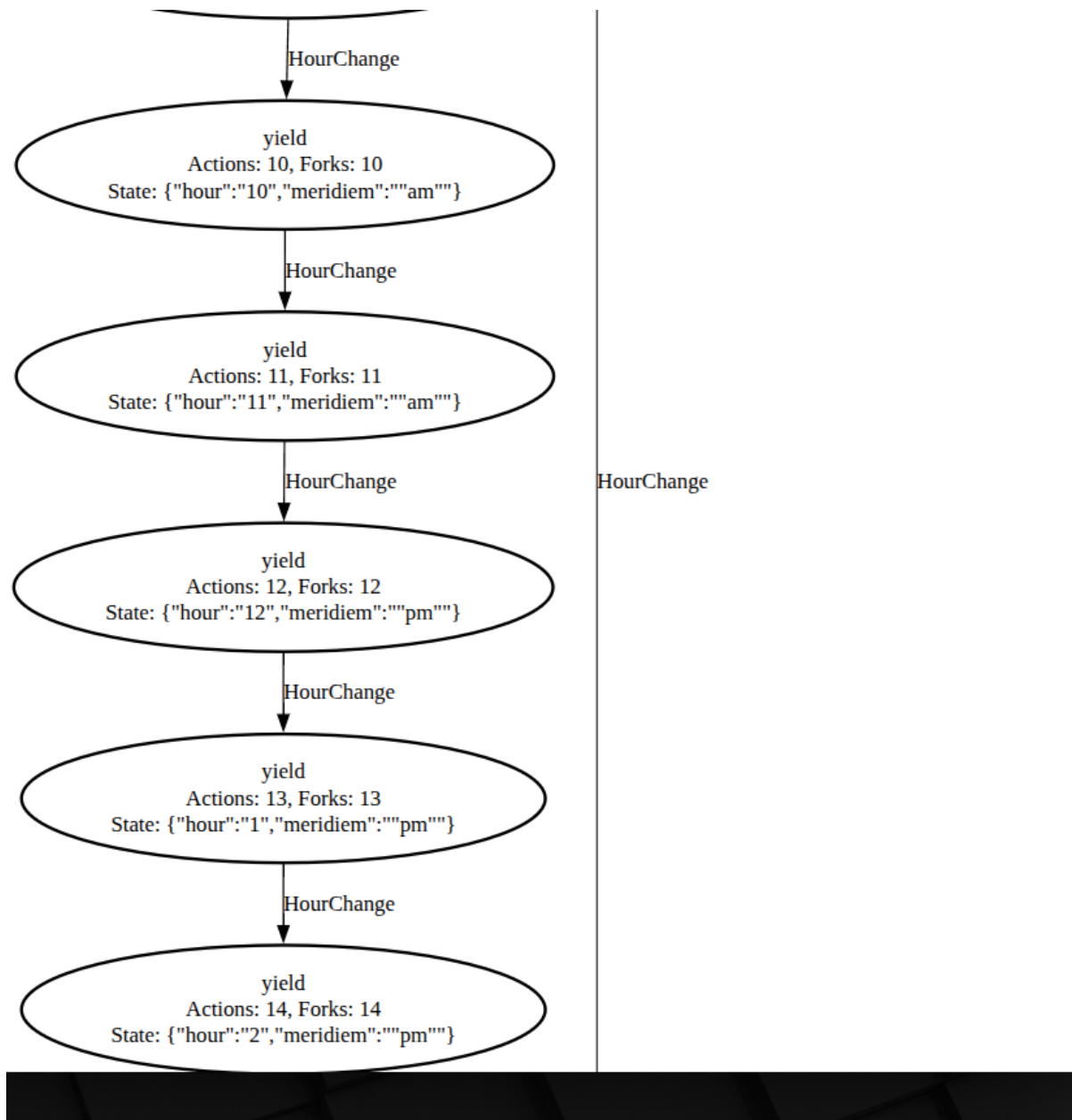
```

Output [States Graph](#)

```

Model checking BuzzyMcBee.json
configFileName: fizz.yaml
StateSpaceOptions: options:{max_actions:100 max_concurrent_actions:2 crash_on_y
Nodes: 24, queued: 0, elapsed: 71.345444ms
Time taken for model checking: 71.395419ms
Valid Nodes: 24 Unique states: 24
IsLive: true
Time taken to check liveness: 161.082158ms
PASSED: Model checker completed successfully

```



Ex 4

```
action Init:
  color = "red"

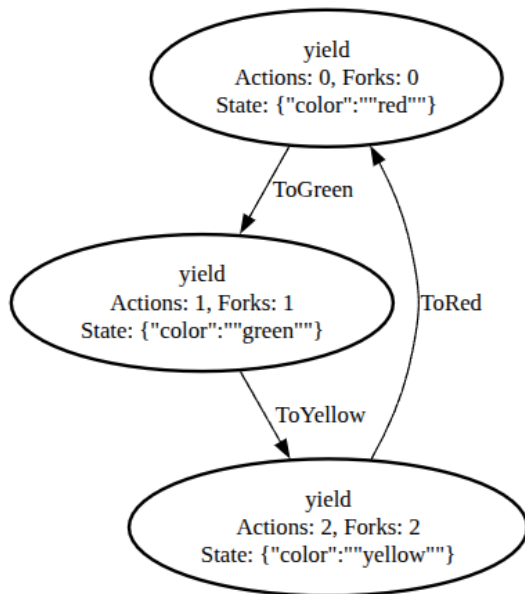
atomic action ToGreen:
  require color == "red"
  color = "green"

atomic action ToYellow:
  require color == "green"
  color = "yellow"

atomic action ToRed:
  require color == "yellow"
  color = "red"
```

Output [States Graph](#)

```
Model checking BuzzyMcBee.json
configFileName: fizz.yaml
StateSpaceOptions: options:{max_actions:100 max_concurrent_actions:2 crash_on_y
Nodes: 3, queued: 0, elapsed: 32.867805ms
Time taken for model checking: 32.910443ms
Valid Nodes: 3 Unique states: 3
IsLive: true
Time taken to check liveness: 133.039687ms
PASSED: Model checker completed successfully
```



Ex 5

```

action Init:
  color = "red"

atomic action ToGreen:
  require color == "red"
  color = "green"

atomic action ToYellow:
  require color == "green"
  color = "yellow"

atomic action ToRed:
  require color == "yellow"
  color = "red"

atomic action Broken:
  require color != "broken"
  color = "broken"

atomic action Fix:
  require color == "broken"
  color = "red"

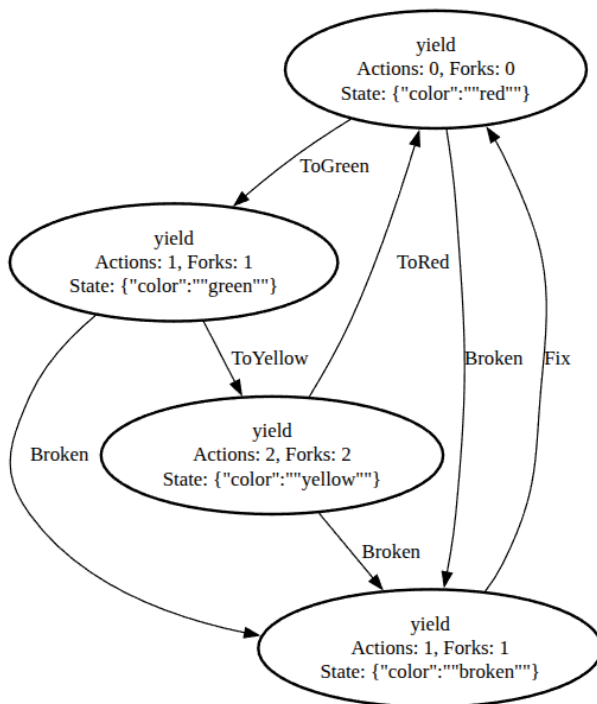
```

Output [States Graph](#)

```

Model checking BuzzyMcBee.json
configFileName: fizz.yaml
StateSpaceOptions: options:{max actions:100 max_concurrent_actions:2 crash_on_y
Nodes: 4, queued: 0, elapsed: 28.061433ms
Time taken for model checking: 28.09864ms
Valid Nodes: 4 Unique states: 4
IsLive: true
Time taken to check liveness: 161.360683ms
PASSED: Model checker completed successfully

```



Ex 6

```

1 always eventually assertion SafelyShutdown:
2   return status == "shutdown"
3
4 action Init:
5   status = "idle"
6
7 atomic action Process:
8   if status == "idle":
9     status = "working"
10
11 atomic action Finish:
12   if status == "working":
13     status = "idle"
14
15 atomic fair<strong> action Shutdown:
16   if status != "working":
17     status = "shutdown"

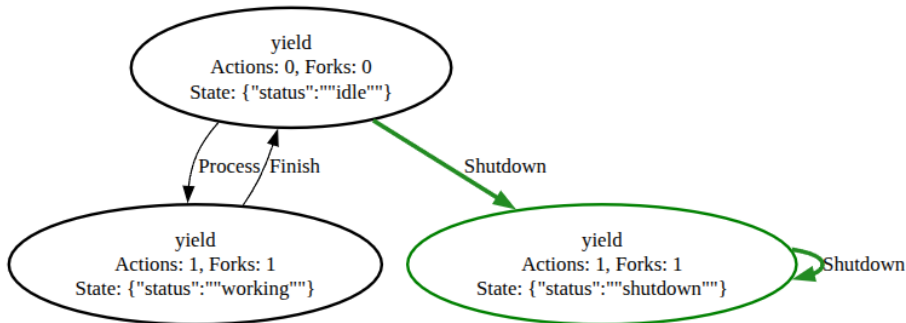
```

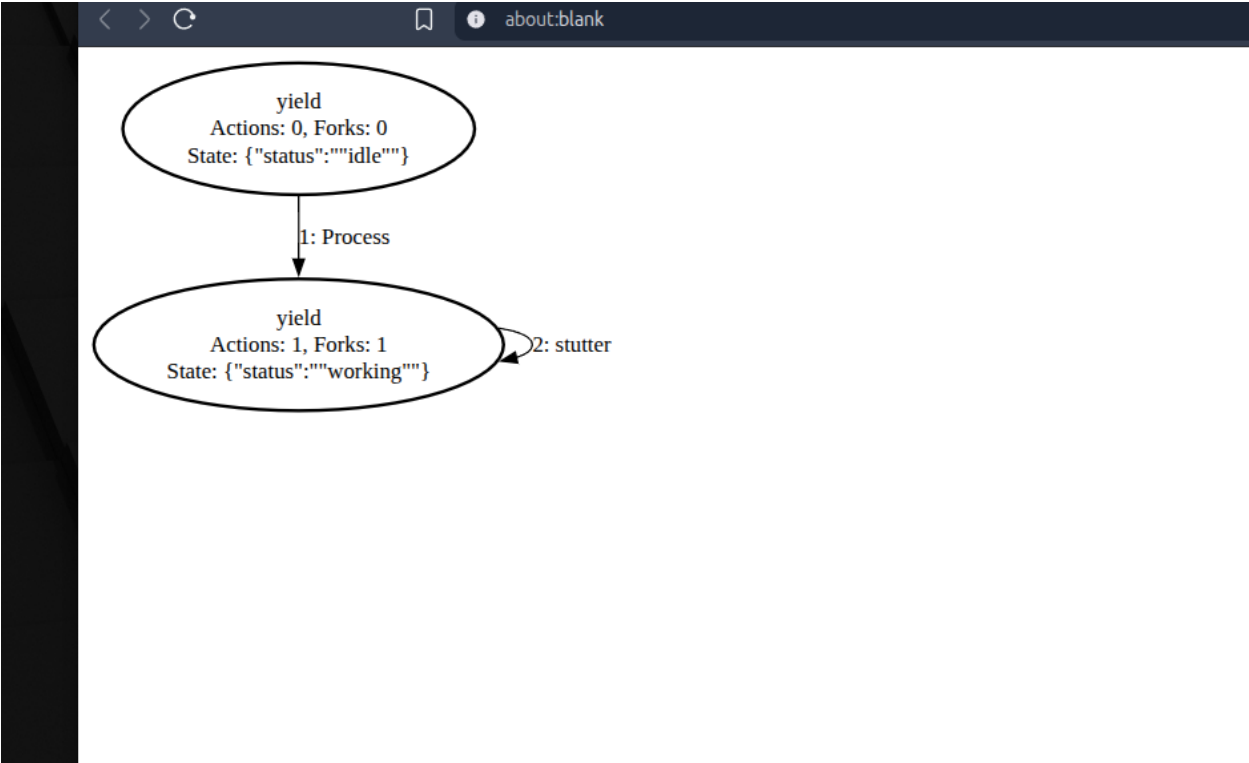
Output [States Graph](#) [Error Graph](#) [Error Formatted](#) [Explorer](#)

```

Model checking BuzzyMcBee.json
configFileName: fizz.yaml
StateSpaceOptions: options:{max_actions:100 max_concurrent_actions:2 crash_on_y
Nodes: 3, queued: 0, elapsed: 28.450541ms
Time taken for model checking: 28.489816ms
Valid Nodes: 3 Unique states: 3
Checking always eventually SafelyShutdown
IsLive: false
Time taken to check liveness: 131.81298ms
FAILED: Liveness check failed
Invariant: SafelyShutdown
-----
Init
--
state: {"status":"idle\"}
-----
Process
--
state: {"status":"working\"}
-----
stutter
--
state: {"status":"working\"}
-----

```





Error States Diff

Row	Thread 0	Yield?	Diff Link	Yield Diff
1	Init	yield		
2	Process	yield	Show diff	Show yield diff
3	stutter	yield	Show diff	Show yield diff

```
1 always eventually assertion SafelyShutdown:
2   return status == "shutdown"
3
4 action Init:
5   status = "idle"
6
7 atomic action Process:
8   if status == "idle":
9     status = "working"
10
11 atomic action Finish:
12   if status == "working":
13     status = "idle"
14
15 atomic fair-strong action Shutdown:
16   if status != "working":
17     status = "shutdown"
```